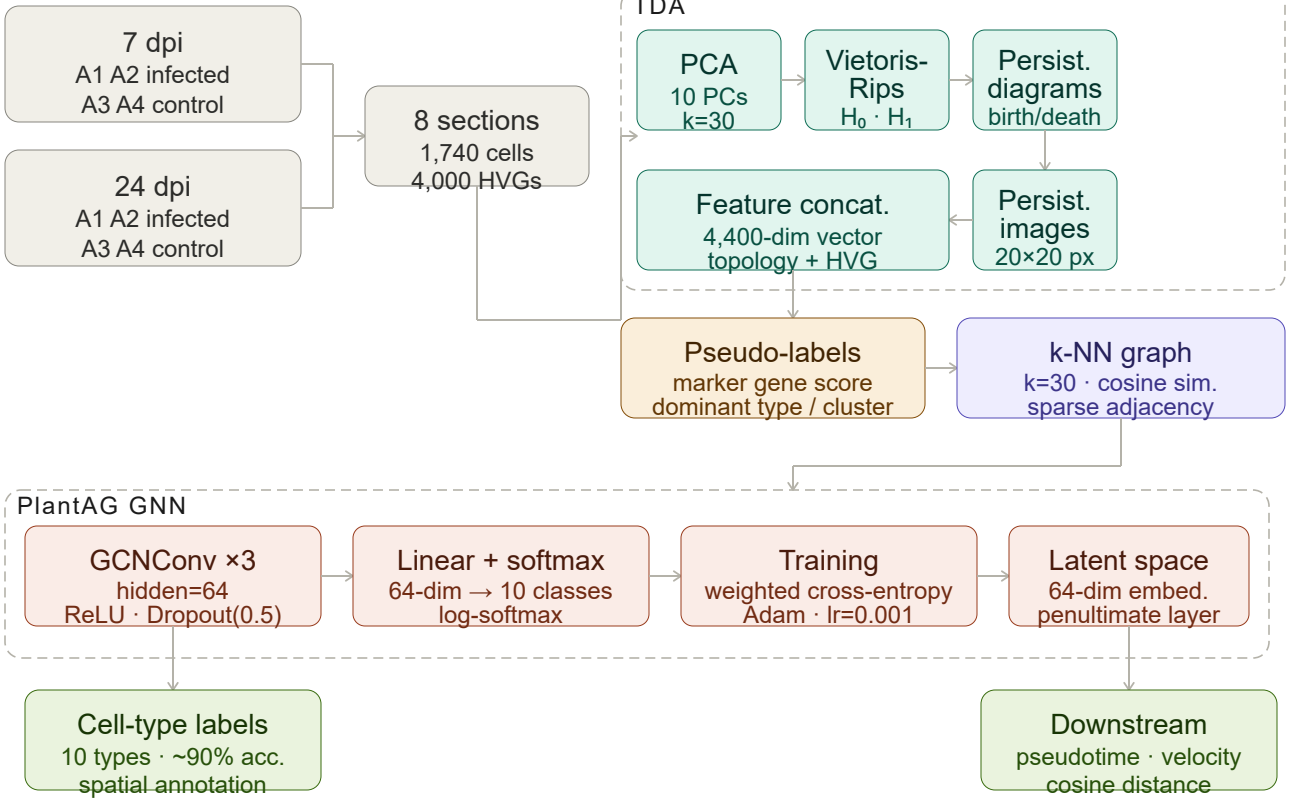
7 dpi
A1 A2 infected
A3 A4 control
24 dpi
A1 A2 infected
A3 A4 control
8 sections
1,740 cells
4,000 HVGs
TDA
PCA
10 PCs
k=30
Vietoris-Rips
$H_0 \cdot H_1$
Persist. diagrams
birth/death
Persist. images
20×20 px
Feature concat.
4,400-dim vector
topology + HVG
Pseudo-labels
marker gene score
dominant type / cluster
k-NN graph
k=30 · cosine sim.
sparse adjacency
PlantAG GNN
GCNConv ×3
hidden=64
ReLU · Dropout(0.5)
Linear + softmax
64-dim → 10 classes
log-softmax
Training
weighted cross-entropy
Adam · lr=0.001
Latent space
64-dim embed.
penultimate layer
Cell-type labels
10 types · ~90% acc.
spatial annotation
Downstream
pseudotime · velocity
cosine distance